

Analyze X-ray or MRI images for disease prediction

**KNEE OSTEOARTHRITIS PREDICTION AND
PROGRESSION USING
MULTI MODAL DEEP LEARNING**

Project No: 25-26J-112

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IT22223708

BSc (Hons) Degree in Information Technology Specialization in
Data Science

Department of Computing

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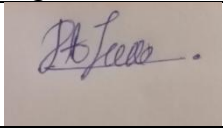
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DECLARATION

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the nonexclusive right to reproduce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books).

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Signature of the Co-Supervisor	Date

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ABSTRACT

Knee Osteoarthritis (KOA) is a progressive degenerative joint disease that leads to pain, reduced mobility, and long-term disability, significantly impacting the quality of life of affected individuals. Early detection and accurate diagnosis are essential for effective treatment and disease management. However, conventional diagnostic approaches such as X-ray imaging primarily capture structural bone changes, while Magnetic Resonance Imaging (MRI) provides detailed information about soft tissues such as cartilage. Relying on a single imaging modality often limits diagnostic accuracy and fails to provide a comprehensive understanding of the disease.

This research presents a deep learning-based dual-modality image classification system for the automated detection of KOA using both X-ray and MRI images. The proposed approach incorporates advanced image preprocessing techniques, including noise reduction, contrast enhancement, and region-of-interest extraction, to improve image quality and focus on the knee joint area. Separate Convolutional Neural Network (CNN) models are developed for each imaging modality, enabling the system to learn modality-specific features. X-rays emphasizing bone structure and joint space narrowing, and MRIs capturing cartilage degradation and soft tissue abnormalities. To enhance diagnostic performance, the system integrates features from both modalities through a multi-modal fusion strategy, allowing complementary information to be combined for improved classification accuracy. The models are trained and evaluated using clinically relevant performance metrics such as accuracy, sensitivity, specificity, and ROC-AUC to ensure robustness and reliability.

Keywords—Knee Osteoarthritis, Multimodal Analysis, X-ray, Clinical Biomarkers, Wearable Monitoring, IoT

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LIST OF ABBREVIATIONS

Abbreviations	Description
AI	Artificial intelligence
CNN	Convolutional neural network
CLAHE	Contrast Limited Adaptive Histogram Equalization
Grad-CAM	Gradient-weighted Class Activation Mapping
IoT	Internet of Things
KL grading	Kellgren and Lawrence grading
KOA	Knee Osteoarthritis
MRI	Magnetic resonance imaging
RCNN	Region-based Convolutional Neural Networks
ROI	Region of Interest
SHAP	SHapley Additive exPlanations
VAG	Vibroarthrography

1 Introduction

Knee Osteoarthritis (KOA) is a chronic degenerative joint disease characterized by the gradual breakdown of cartilage, leading to pain, stiffness, and reduced mobility. It is one of the most common musculoskeletal disorders worldwide and significantly impacts the quality of life of affected individuals, particularly among the aging population [1]. As the disease progresses, structural changes occur within the knee joint, including joint space narrowing, osteophyte formation, and subchondral bone sclerosis, which ultimately result in functional limitations and disability.

Traditionally, the diagnosis and assessment of KOA rely on clinical evaluation and medical imaging techniques. Among these, radiographic imaging using X-rays remains the most widely used diagnostic method due to its accessibility and cost-effectiveness. The Kellgren–Lawrence (KL) grading system is commonly used to classify the severity of KOA based on radiographic features [2]. However, X-ray imaging primarily captures bone-related abnormalities and often fails to detect early-stage cartilage degeneration, limiting its effectiveness for early diagnosis.



Figure 1 Knee X-ray of a normal person



Figure 2 Knee X-ray of a KOA patient

Magnetic Resonance Imaging (MRI) provides a more comprehensive view of the knee joint by capturing both bone and soft tissue structures, including cartilage, ligaments, and synovial tissues [3]. This enables the detection of early degenerative changes that are not visible in X-ray images. Despite its advantages, MRI is less frequently used in routine diagnosis due to higher costs and limited accessibility, especially in low-resource healthcare settings.

Recent advancements in artificial intelligence, particularly deep learning, have significantly improved the analysis of medical images. Convolutional Neural Networks (CNNs) have demonstrated strong capabilities in automatically extracting features and performing classification tasks for KOA detection [4], [5]. Various studies have applied deep learning models to X-ray images for automated KOA classification, achieving promising results in terms of accuracy and reliability [6]. Similarly, machine

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learning techniques have been applied to MRI data to predict disease progression and identify subtle soft tissue abnormalities [7], [8].

However, most existing approaches rely on a single imaging modality, either X-ray or MRI, which limits the overall diagnostic capability. Since KOA affects both bone and soft tissue structures, relying on a single modality provides only a partial understanding of the disease. Recent research has explored multi-modal approaches, combining different data sources to improve prediction accuracy and provide a more comprehensive diagnosis [9], [10]. These approaches highlight the importance of integrating complementary information from multiple modalities.



Figure 3 MRI of a healthy knee



Figure 4 Knee MRI of a KOA patient

Another critical challenge in AI-based medical systems is the lack of interpretability. Many deep learning models operate as black-box systems, making it difficult for clinicians to understand how predictions are generated. This lack of transparency reduces trust and limits adoption in real-world healthcare environments [13]. To address this issue, explainable AI techniques such as SHAP and Grad-CAM have been introduced to provide visual and quantitative explanations for model predictions [11], [12].

Despite these advancements, challenges remain in developing systems that are accurate, interpretable, and capable of integrating multiple data modalities. This research aims to address these challenges by developing a dual-modality deep learning-based system that analyzes both X-ray and MRI images for KOA detection. By leveraging the strengths of both imaging modalities and incorporating explainable AI techniques, the proposed system seeks to improve diagnostic accuracy, enhance interpretability, and support clinical decision-making.

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1.1 Background Literature

Knee Osteoarthritis (KOA) is a common degenerative joint disease that involves the progressive loss of articular cartilage, changes in the underlying subchondral bone, and inflammation of the joint. This condition is a leading cause of disability, especially in older adults, and can severely impact every aspect of life, including mobility, function and well-being [1]. The rising incidence of KOA is linked to several factors including age, weight, past trauma and lifestyle choices, which highlights its importance as a public health issue.

KOA is typically diagnosed and evaluated through clinical examinations and imaging studies. In particular, X-rays are the most frequently used imaging technique due to their availability and affordability. Often, the Kellgren-Lawrence (KL) grading is used to assess the severity of KOA based on the presence of radiographic signs such as joint space narrowing, osteophytes and bone sclerosis [2]. But X-rays mainly show bone changes and may not show early signs of cartilage degeneration, which is an important factor in disease progression.

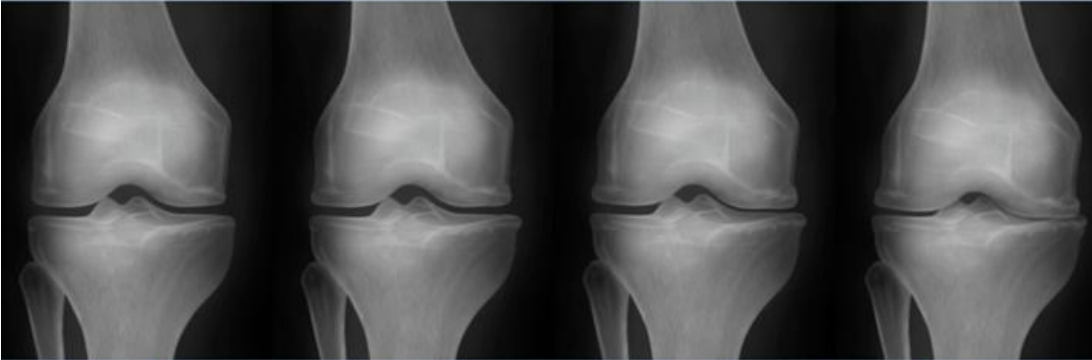
Kellgren-Lawrence (KL) grading scale					
					
	Grade 1	Grade 2	Grade 3	Grade 4	
CLASSIFICATION	Normal	Doubtful	Mild	Moderate	Severe
DESCRIPTION	No features of OA	Minute osteophyte: doubtful significance	Definite osteophyte: normal joint space	Moderate joint space reduction	Joint space greatly reduced: subchondral sclerosis

Figure 5 KL Grading Scale

Magnetic Resonance Imaging (MRI) is a more sophisticated imaging technique that offers greater clarity of bone as well as soft tissues such as cartilage, ligaments and synovial tissues. MRI allows detection of early signs of disease not evident on X-rays, which can aid early and accurate diagnosis of KOA [3]. However, MRI is not widely used in routine clinical settings because of its relatively high cost, longer acquisition time and lack of accessibility, particularly in resource-limited settings.

The advent of Artificial Intelligence (AI) technologies, especially deep learning, has revolutionized medical image processing. Convolutional Neural Networks (CNNs)

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have proven to be particularly effective in automatically learning features from medical images and classifying them with high accuracy. Several studies have demonstrated the ability of CNN models to accurately detect and classify severity levels of KOA from X-ray images with performance on par with experts [4], [5]. Also, region-based detection and specific architecture optimizations have improved detection accuracy using regions of interest in knee images [6].

Likewise, deep learning models are also used to predict the progression of KOA and to identify soft tissue changes in MRI images. MRI-based machine learning has demonstrated high sensitivity to early signs of cartilage degeneration and disease-associated structural changes [7]. Recent improvements in image segmentation and feature extraction have also improved the ability of MRI-based models to provide detailed anatomical features [8].

So far, such techniques have been successful, but most systems are unimodal and rely on either X-rays or MRIs. This restricts the ability of the systems to make an accurate diagnosis because each imaging modality reveals only a subset of disease information. X-rays are good at detecting bone lesions, while MRIs reveal soft tissue abnormalities. As a consequence, this can result in a decrease in accuracy and a lack of generalization to real-world scenarios.

To overcome this challenge, recent studies have investigated multi-modal learning methods that combine different data sources. Research has demonstrated the benefits of integrating multiple imaging modalities (such as X-ray and MRI) and clinical information to enhance predictive accuracy and gain a holistic view of KOA [9], [10]. Multi-modal fusion approaches allow models to use the additional information from different modalities, resulting in improved diagnostic performance and stability.

Additionally, the incorporation of explainable AI (XAI) approaches is crucial for medical applications [13]. Techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP) provide visual and quantitative explanations for model decisions, aiding clinicians in interpreting the basis for automated diagnostic decision-making [11], [12]. Such explanations are crucial for gaining confidence and promoting the use of AI-based tools in medical practice.

Overall, the current literature underscores the increasing role of deep learning and multi-modality in the detection and diagnosis of KOA. While advances are being made, there are challenges in developing models that leverage multiple imaging techniques while maintaining accuracy, interpretability and clinical relevance. These gaps highlight the need for a holistic dual-modality approach that leverages X-ray and MRI imaging for better and early detection of Knee Osteoarthritis.

1.2 Research gap

Knee osteoarthritis (KOA) is a major public health issue with a progressive nature and a hard-to-diagnose process for early detection and ongoing monitoring. While many studies have attempted to automate detection and assessment of severity for OA using medical imaging and machine learning, there are important gaps in current research.

One significant gap in the literature is the use of unimodal approaches, which include the use of X-ray images, MRI scans or structured clinical data for model training. While X-ray-based research has shown promise in detecting structural changes using deep learning, it fails to detect changes in early cartilage degeneration [4], [5]. Alternatively, studies based on MRI offer a better understanding of soft tissue structures, but are not typically combined with X-rays, and consequently lead to disjointed diagnostic tools [7]. This limits ability to understand KOA because the disease affects both bone and cartilage.

Moreover, while some recent research has started to explore multi-modal learning, the combination of multimodal data is still not fully integrated. For instance, some studies integrate imaging and clinical data, but fail to leverage the combined information and insights from X-ray and MRI imagery [10]. As shown in Table , most current approaches do not deliver a full integration of bone structure (X-ray) analysis, soft tissue (MRI) analysis, feature extraction through deep learning techniques, and integration of multi-modal insights in a unified manner. Such a failure of complete fusion reduces the accuracy and clinical relevance of the prediction systems.

An additional shortcoming is the lack of integration of explainable AI (XAI) into KOA prediction models. Deep learning models may be highly accurate but they can be regarded as "black boxes", which make predictions without explanations. In clinical settings, this opacity leads to lower trust among health practitioners and limits application in clinical practice [13]. While interpretability methods like Grad-CAM and SHAP have been proposed, their use in KOA detection is still not common [11], [12].

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Moreover, many studies are focused only on static data collected at one point of time (cross-sectional) and not on the progressive nature of the disease. In practice, KOA progression needs to be monitored and analysed over time, which most models do not consider. This impacts the usefulness of such systems in predicting future disease progressions and early intervention approaches.

Further, current approaches are often not general enough or scalable, given they are developed and evaluated using research data in ideal clinical settings. This limits their use in varied clinical settings, especially in resource-poor settings with limited access to high-technology diagnostic methods.

Product	Focus on bone structure (X-ray)	Focus on soft tissue/cartilage (MRI)	Deep learning-based feature extraction	Fusion of X-ray & MRI insights	Use XAI
Tiulpin et al [10]	✓	x	✓	x	x
Liu et al [8]	x	✓	✓	x	x
Selvaraju et al [12]	x	x	x	x	✓
Chen et al 2025 [14]	x	x	✓	x	✓
Proposed Solution	✓	✓	✓	✓	✓

Table 1 Comparison of former researches

The key research gaps identified are

1. Absence of multimodal (X-ray and MRI) integrated systems,
2. The lack of explainable AI techniques to increase model interpretability,
3. The lack of extendable and clinically relevant systems that can facilitate early diagnosis and ongoing monitoring.

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These considerations suggest the need for a comprehensive, explainable and multi-modal deep learning framework that integrates X-ray and MRI imaging to improve the accuracy and effectiveness of KOA diagnosis.

1.3 Research Problem

Osteoarthritis of the Knee (KOA) is a chronic degenerative joint condition that affects mobility and quality of life. Timely diagnosis and proper detection are critical for treatment and to avoid severe complications. But current methods of diagnosis are still lacking in comprehensively evaluating the disease.

The conventional clinical diagnosis approach mainly relies on radiography, using X-rays and the Kellgren-Lawrence (KL) classification, which investigates bone changes, such as joint space narrowing and osteophyte formation [2]. While this approach is commonly employed, it cannot capture early signs of cartilage damage, which play a crucial role in the development of KOA. In contrast, Magnetic Resonance Imaging (MRI) offers a more comprehensive view of soft tissues, such as cartilage and ligaments, for early detection of changes [3]. Despite this, MRI diagnosis is used solely and is not well integrated with X-rays in most clinical or computational approaches.

With recent breakthroughs in deep learning, automated KOA diagnosis from medical images has become possible. CNNs have proved to be effective in diagnosing KOA from X-rays and MRI scans [4], [7]. But such methods are unimodal, i.e., they only use one medical image at a time, and thus lack information about the full pathology of the condition. This can lead to lower accuracy and less clinical applicability.

The lack of interpretability is another significant issue impacting AI-based systems. Deep learning models are often black-box models, which only provide predictions without revealing the reasoning used to arrive at their conclusions. This hampers clinician trust and the uptake of these systems in the clinical environment [13]. Deliberations of models in high-stakes medical scenarios also lack interpretability.

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There is no holistic, robust, accurate and explainable deep learning model that effectively combines these two sets of X-ray and MRI images to predict the presence of Knee Osteoarthritis (KOA) and takes advantage of the synergy between the two imaging modalities.

To tackle this issue, the study proposes to create a bi-modality image classification system that integrates X-ray and MRI data using a deep learning architecture. The system needs to not only enhance detection performance but also provide robustness, clinical appropriateness and interpretability, thereby enabling early diagnosis and help with decision making for the treatment of KOA.

1.3 Research Objectives

1.3.1 Main Objective

This study seeks to build a precise and robust deep learning-based system that can accurately detect KOA from medical images. The research addresses a class of learning techniques that can exploit both X-rays and MRI for enhanced diagnosis and early detection of KOA. The system combines state-of-the-art image processing and deep learning algorithms to deliver a clinically meaningful and scalable KOA detection system.

1.3.2 Sub-Objectives

1. To acquire and validate image datasets.
 - Collect X-ray and MRI data from public sources and clinical settings, with validation for quality and accuracy by experts.
2. To enhance the images.
 - Use noise reduction, contrast enhancement, normalization and Region of Interest (ROI) extraction to enhance images and isolate the knee joint.
3. To create separate deep learning models for each modality.
 - Develop and train Convolutional Neural Network (CNN) models for X-ray and MRI images, respectively, to learn modality-specific characteristics related to bone and soft tissue health.
4. To use transfer learning to enhance feature learning.

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- Make use of pre-trained deep learning models (e.g., ResNet, DenseNet, EfficientNet) to improve model performance and training efficiency.
5. To assess model performance with clinical metrics.
 - Evaluate the quality of the proposed models by the metrics of accuracy, sensitivity, specificity, precision, F1-score, and ROC-AUC.
 6. To aid in the diagnosis of early Knee Osteoarthritis.
 - Make the system able to detect KOA at an early stage using the complementarity of the two imaging techniques.
 7. To improve its accuracy and scalability for clinical use.
 - Create a system that will be reliable, scalable, and deployable in health care systems for medical diagnosis.

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2 Methodology

2.1 Methodology

2.1.1 System Overview

The system is a multi-modal intelligent system for predicting, monitoring and grading KOA using multiple data sources such as medical images, clinical and biomarker data, and real-time IoT data. The system architecture, as depicted in the diagram below, is comprised of four functional modules, modeling of clinical and biomarker data, prediction of the disease based on multi-modality images, grading of KOA severity and monitoring of knee health using IoT signals.

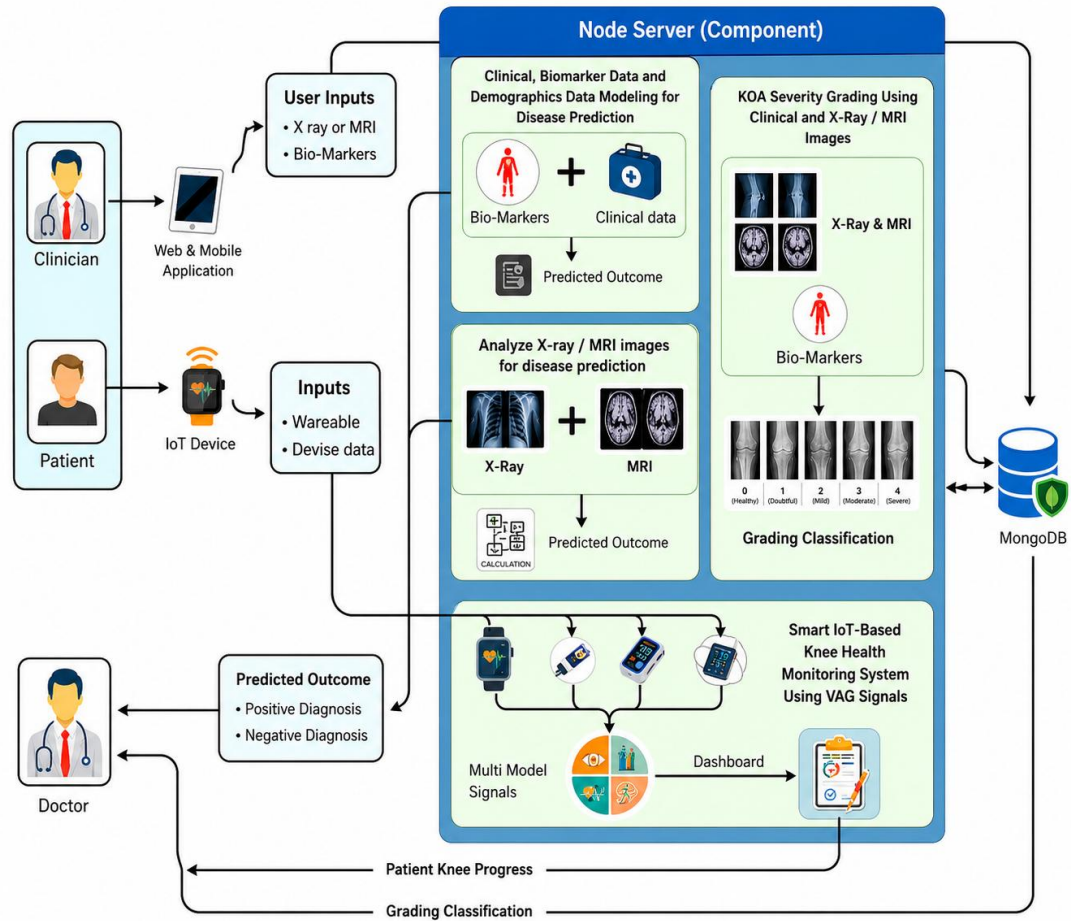


Figure 6 Overall System Diagram

In the first stage, data is gathered from various sources via a web-mobile based portal. The system receives data from clinicians or patients, including X-ray and MRI images, and biomarkers or clinical data. Simultaneously, IoT-based wearable gadgets keep

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track of the knee joint signals during everyday activities. Data from all sources are sent to a central processing node (Node Server) for all analysis operations.

The module for clinical data, biomarker, and demographic data modeling concentrates on the structured data to identify the risk of KOA. This component produces a risk outcome using patient-specific data such as age, body mass index, and medical and laboratory data. It employs machine learning algorithms to detect relationships between the risk factors and the disease.

The KOA severity grading module uses both image and structured data to determine the severity of KOA using various grading scales like the Kellgren-Lawrence scale. This module integrates the results of image-based models and clinical data models to give a full picture of the severity of the disease from healthy to severe KOA.

The IoT-enabled knee health monitoring component leverages wearable technologies to obtain real-time Vibroarthrography (VAG) signals and other biomechanical implications. These signals provide insights into joint movement and mechanical abnormalities, enabling continuous monitoring of disease progression. The information is represented visually in an interactive dashboard, enabling clinicians and patients to monitor the health status of the knee joint and be alerted in the presence of any abnormality.

The main focus of this work is the multi-modality image-based component for predicting disease, using X-ray and MRI images to determine whether KOA is present. It has a systematic approach, as shown in the component diagram. First, medical images are preprocessed by denoising, contrast adjustment, normalization, and Region of Interest (ROI) extraction to enhance image quality and focus on the knee joint. The processed images are used to develop deep learning models. Two independent Convolutional Neural Network (CNN) models are trained for X-ray and MRI images to learn their respective features. X-ray images are primarily used to detect bone and joint space narrowing, as well as osteophytes, whereas MRI images are key to detecting soft tissue variations, such as cartilage loss and ligament tears. This approach allows the system to learn the unique characteristics of each modality.

Training models using data augmentation (rotation, flipping) helps to increase model performance. The models are assessed using cross-validation and various performance measures such as accuracy, F1-score, and ROC-AUC. The predictions of the two models are then merged to make a final diagnosis of the presence or absence of KOA.

Moreover, to overcome the limitations of black-box models, explainable AI (XAI) methods are used in the system. These techniques offer visual interpretation and insights into the predictions, allowing clinicians to comprehend model reasoning, and help build trust in the system. The results from each component are collected in a database, allowing data to be tracked over time. The system provides the final diagnosis outcome, KOA severity, and disease progression as a user-friendly web dashboard and mobile app. This combined system offers support for both health

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professionals and patients through reliable prediction, tracking patient data, and providing insights for managing KOA.

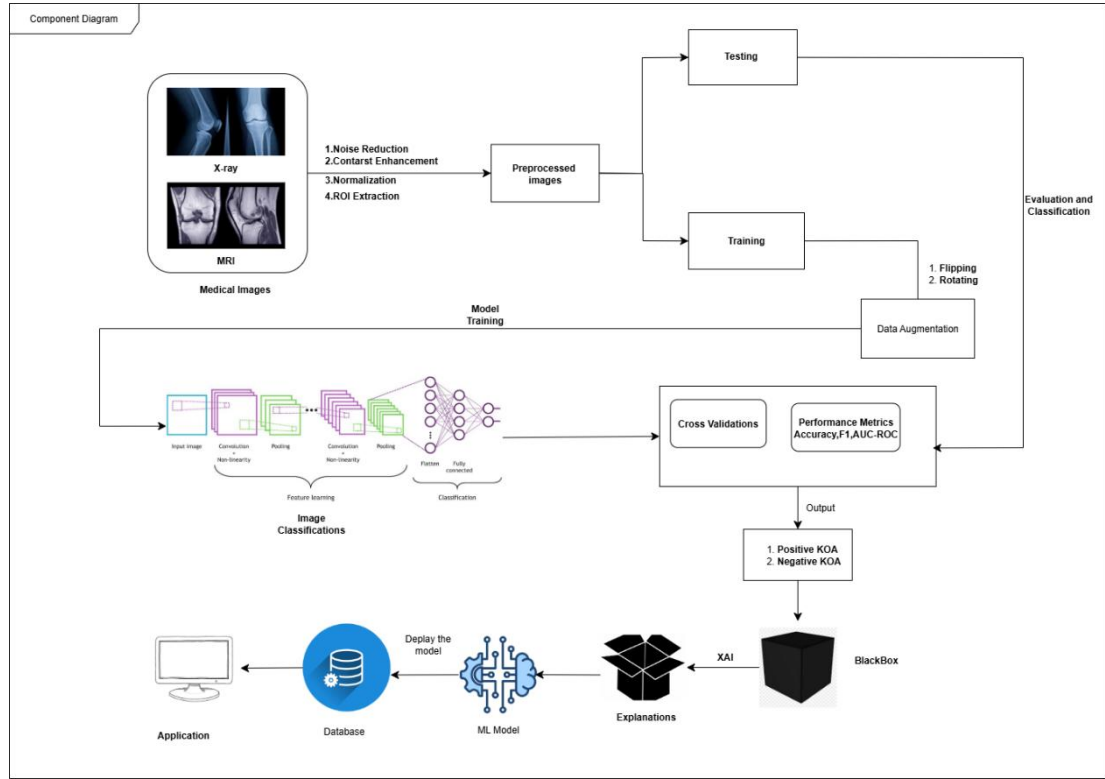


Figure 7 Component Diagram

2.1.2 Data collection and Preprocessing

The success of a deep learning system for medical imaging applications depends on the quality, variety and pre-processing of the data. In this study, a series of procedures were implemented for data collection and preprocessing to ensure optimal input data for model development and KOA prediction.

Data Collection

The data set in this study included X-rays and MR images of knee joints obtained from publicly available medical imaging databases and research data sets. This contains images from both positive and negative cases of KOA, allowing for the construction of binary classification models to diagnose the disease. Furthermore, the format of the dataset follows clinical standards, ensuring the training of the model on clinically relevant data.

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The data collected ensure reliable and clinically relevant information since X-ray images represent bone abnormalities such as joint space narrowing and osteophyte, while MRI images represent soft tissues such as cartilage degradation and ligament status [3]. This multi-modality approach aids in the understanding of KOA.

The datasets were properly structured in training, validation and testing directories. Each image was correctly labelled into different classes (KOA: positive and negative). This ensured effective training and testing of models.

In this research experimentation, several deep learning models were tested for X-ray image classification, experimenting with various architectures and approaches. Such as LeNet-5, AlexNet, GoogleLeNet, ResNet50, DenseNet201, EfficientNetB3 and MobileNetV3 several models have been implemented in this study. Also, detection models like YOLOv8 and YOLOv11 were considered. Following a comparative study, the YOLO-based model was chosen as the final model for the X-ray-based KOA prediction system due to its excellent performance and efficiency.

For MRI-based KOA prediction, a YOLO-based model was also used and trained with a customised implementation. This YOLOv8 model was chosen as the final model for MRI-based KOA detection as it was able to capture pertinent features and classify the KOA state into the five KOA grades.

Data Preprocessing

Medical images can be prone to noise, different lighting conditions and clutter, which can impact on model performance. As a result, a series of preprocessing techniques were applied to X-rays and MRI scans to improve image clarity and facilitate learning.

1. Noise Reduction
 - Images were filtered using techniques like Gaussian filtering and median filtering to reduce noise while maintaining structural integrity. This helps to avoid distraction from important features.
2. Contrast Enhancement
 - Contrast Limited Adaptive Histogram Equalization (CLAHE) was used to improve contrast and enhance important features and regions of cartilage and joint boundaries. This enhances the features of key anatomical structures, especially in low-contrast X-rays.
3. Normalization
 - The images were normalized to a consistent range of pixel values (usually 0-1 or -1 to 1) to have a standardized dataset. This ensures stable learning and convergence of the model.
4. Region of Interest (ROI) Extraction

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- The area of the knee joint was cropped to remove any unnecessary information. Isolating the region of interest helps to limit the attention of the network to the relevant region, thus improving the classification performance. In certain cases, the knee area was detected prior to cropping.
5. Image Resizing and Standardization
 - Images were resized to a constant input size for deep learning algorithms. This is necessary for compatibility with neural networks and enables batch processing during model training.
 6. Data Augmentation
 - To enhance generalization and prevent overfitting, data augmentation was performed, including rotation, flipping and minor deformations. These methods mimic natural variations in medical images, and help diversify the dataset without collecting new data.

Training Preparation and Validation

Following data preparation, the dataset was divided into training, validation and test sets for model performance assessment. Cross-validation was used to assess the models' ability to perform on new data.

The preprocessed images were then fed into deep learning models, which automatically extracted features and classified the images. Model performance was assessed using accuracy, F1-score and ROC-AUC.

2.1.3 Functional, Non-Functional Requirements

Functional Requirements

Functional requirements define the key functions of the proposed dual-modality (X-ray and MRI) image-based KOA prediction component, which will focus on using X-ray and MRI images for detection.

1. Data Input and Acquisition
 - The system is intended to take medical images as the input data for the detection of KOA. Clinicians or system administrators can upload X-ray and MRI images via a web portal using commonly used image formats like JPEG and PNG. To ensure quality and consistency, the system validates these images based on imaging modality (X-ray or MRI), resolution and file format. This ensures that the model is not

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misled by erroneous or poor-quality data and only relevant medical images are used for prediction.

2. Image Preprocessing Module.

- Input images are preprocessed to enhance quality and relevance before being used for model training or prediction. This system automatically smooths the images to reduce unwanted artifacts while preserving critical features. The images are also enhanced to increase the contrast of important features in the knee joint, using techniques like histogram equalization. The images are retcalibrated to ensure uniform distribution of the dataset. They are then cropped to focus on the region of interest (ROI), specifically the knee joint area, to remove unnecessary background. Each image is, ultimately, resized and normalised to meet the specifications for input into deep learning models, enabling consistency and efficient processing.

3. Deep Learning Image Processing.

- The main attribute of the component is its capability to perform deep learning-based medical image analysis. The device employs trained YOLO-based models for X-ray and MRI image analysis respectively. The X-ray model is trained to identify structural changes like reduced joint space and bone deformities, while the MRI model is trained to learn soft tissue-related features such as cartilage degeneration and ligament issues. Both models output predictions of KOA positive or negative. The combined results from both models are used to make a more-confident final prediction, taking advantage of complementary features of X-ray and MRI images.

4. Prediction Output Generation

- The system generates an easy-to-understand prediction output from the medical images. The output is a decision of KOA-positive or KOA-negative. Furthermore, the system offers a confidence score for the prediction, enabling the assessment of prediction trustworthiness. This can be used as an aid to the clinician to provide rapid and automated diagnostic information.

5. Explainable AI (XAI)

- To improve the transparency and confidence in the model, explainable AI (XAI) methods are integrated into the forecasting process. The system produces visual explanations (e.g., heatmaps) that show the parts of the X-rays or MRIs that were most important for the model to make its prediction. This feature helps clinicians interpret the model's

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reasoning process, linking the model's predictions with clinical knowledge. This allows the system to both predict and support the interpretability and validation of the output.

6. Model Evaluation and Validation

- The system supports evaluation of the models using standard measures such as accuracy, F1-score and ROC-AUC. Use cross-validation during model training to improve the models' ability to perform on new data. This assessment is essential for ensuring the system's accuracy and stability, which ensures that the predictions are consistent across different data sets and environments.

Non-Functional Requirements

Non-functional requirements specify the non-functional properties of the image-based KOA prediction system.

1. Accuracy and Reliability

- The system aims to deliver high diagnostic accuracy through the use of fine-tuned deep learning models and properly preprocessed data. Particular attention is given to reduce the false negatives, as early diagnosis is essential to the treatment of KOA. The system should be robust and consistent in terms of performance across different imaging settings and data, making it a viable solution for medical practice.

2. Performance Efficiency

- The system is expected to be able to make predictions within a reasonable amount of time for real-time or near real-time applications. The component can provide efficient processing by using efficient model architectures and computational hardware (such as GPU acceleration). This allows the processing of a large medical image workload without excessive delay.

3. Scalability

- It is scalable, capable of processing a growing number of medical images. It can accommodate further growth, such as the addition of other types of imaging or more advanced deep learning models. This allows the system to keep up with developments in medical imaging and artificial intelligence.

4. Usability

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- The system is designed with usability in mind. The system will be designed to have a user-friendly interface, enabling clinicians to easily upload images and interpret the results, with a minimum of technical skill required. The results are displayed in an easily digestible manner, allowing users to understand the predictions and explanations without being experts in machine learning.

5. Security and Data Privacy

- Health data is highly confidential and the system takes measures to protect patient data. Medical images are securely stored and transmitted, and the system provides authentication to control access. Privacy protocols are in place to safeguard patient privacy and adhere to medical regulations.

6. Maintainability

- The system is built with a modular design, enabling changes or enhancements to different parts of the system without impacting the entire system. This allows for simple maintenance, retraining of models, and upgrades to the system as new data or models are developed. This ensures the system's sustainability over time.

7. Interpretability and Trust

- For clinical use the system focuses on explainability and interpretability. Through the use of explainable AI and confidence scores, the system allows users to comprehend and trust the generated recommendations. This is essential in healthcare to support decision-making with reliable evidence.

2.2 Commercialization Aspect of the Product

The dual-modality KOA detection system has considerable commercial potential because it offers precise, rapid and affordable diagnostic assistance using non-expensive and widely accessible medical imaging methods. The system uses deep learning to process X-ray and MRI scans to overcome key problems in early diagnosis and efficiency of diagnosis, making it applicable in the clinical environment.

2.2.1 Market Opportunity

Osteoarthritis of the knee is a common musculoskeletal condition, especially in the elderly. The need for early diagnosis and ongoing monitoring has led to a need for smart diagnostic systems to help clinicians make quicker and better diagnostic

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decisions. Conventional procedures involve expert interpretation and lengthy processes, which may hinder its efficiency in busy clinics.

The solution can be marketed as a decision support system that supports clinicians by automating the diagnosis of KOA from medical images. The primary market for the system is hospitals, diagnostic imaging facilities, and radiology departments, where it can be incorporated into the current imaging process. The system is also particularly valuable in low-resource settings, where access to specialist radiologists might be scarce. The system offers automatic analysis, which helps overcome the shortage of skilled medical staff.

2.2.2 Value Proposition

The significance of the system is the use of two-modality analysis (X-ray and MRI) for more accurate diagnosis. This system takes into account complementary information from bone analysis and soft tissue analysis, unlike traditional systems using a single-modality approach. This leads to improved prediction accuracy and facilitates the detection of KOA in its early stages.

Moreover, the use of deep learning models based on YOLO facilitates fast and scalable processing of images, making it a viable choice for real-time or near real-time scenarios. The system's explainable AI (XAI) capabilities improve its interpretability by visually explaining predictions, which is essential for clinician acceptance. This blend of performance, speed, and explainability sets the system apart from current diagnostic systems.

2.2.3 Deployment Strategy

The solution can be implemented either on-premise or in the cloud, depending on healthcare institution's needs. A cloud-based system allows for secure upload of medical images, analysis by centrally hosted AI systems and display of the results in web or mobile apps. This provides scalability and remote access.

On the other hand, the system can be embedded in a hospital's IT system as an on-site AI-assisted diagnostic tool, providing a quicker turnaround and adhering to data protection laws. The system's modular design enables easy integration with hospital Picture Archiving and Communication Systems (PACS) and Electronic Health Record (EHR) systems, facilitating its clinical adoption.

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2.2.4 Business Model

A viable approach for commercialization of the system is to adopt a Software-as-a-Service (SaaS) business model, in which healthcare facilities pay on a subscription basis, either per transaction or per user. This guarantees ongoing support, maintenance and upgrades for the system.

Alternatively, a licensing model can be used for large hospitals and diagnostic laboratories who want to deploy the system on-premise. Enhanced features like analytics, reporting and interoperability with other systems can be offered as part of the system.

To increase uptake, particularly in emerging markets, a low-cost or subscription-based model can be offered to enable smaller clinics and rural health centers to afford the system. This enhances the system's reach without compromising profitability.

2.2.5 Competitive Advantage

This system has a number of advantages. The incorporation of both X-ray and MRI data in a single deep learning framework offers a holistic approach to diagnosis, unlike single-modality systems. By incorporating YOLO models, the system is capable of rapid processing and feature extraction, enabling real-time applications.

Furthermore, the potential use of explainable AI techniques improves interpretability, which is often absent in traditional AI systems used in diagnosis. This increases trust and facilitates regulatory approval processes. Moreover, the scalability, modularity and integration characteristics of the system provide further advantages as a feasible health care solution.

2.3 Testing and Implementation

2.3.1 Implementation

The implementation of the proposed bimodal Knee Osteoarthritis (KOA) diagnostic system involves the design and training of deep learning models for the classification of KOA from X-rays and MRI scans. Implementing the proposed system involves a series of steps: model experimentation, training, evaluation, and model selection.

Development Environment and Tools

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The implementation was conducted in Python, the predominant language for machine learning and medical imaging. Used Jupyter Notebook for development, experiments and result visualizations.

Implemented deep learning models using the latest frameworks such as PyTorch and the Ultralytics YOLO framework, which offers efficient tools for training and deploying object detection and classification models. Additional libraries like OpenCV were employed for image pre-processing, and NumPy and Matplotlib were used for data manipulation and visualization. GPU acceleration was used to speed up the model training process, allowing for the effective processing of sizable image datasets and deep learning models.

X-ray Image Model Development

Several experimental implementations of deep learning models were conducted to determine the best model for detecting KOA in X-ray images. These experiments were run in several Jupyter Notebook files. This made it possible to compare classical Convolutional Neural Network (CNN) models such as LeNet-5, AlexNet, GoogLeNet, ResNet50, DenseNet201, EfficientNetB3 and MobileNetV3, with advanced models such as YOLO-based models. Each model was trained on preprocessed X-ray data and evaluated using metrics, including accuracy, F1-score and ROC-AUC.

Based on experimentation and comparison of different models, it was found that although CNN models were able to extract features from the image, they struggled to capture long spatial information. However, the YOLO model excelled because it was able to do real-time feature detection and localization and was better suited for medical image processing.

Therefore, the YOLO model was used as the final model for KOA detection using X-rays. This model offered better detection accuracy, speed, and better performance on different datasets.

MRI Image Model Development

To detect KOA using MRI, built and trained a model that was specifically developed for MRI. MRI images provide more intricate information about soft tissues, and models need to learn more subtle features, such as cartilage and ligament abnormalities. The YOLO model was chosen for MRI analysis as it can identify regions of interest in images and learn the spatial context. The model was trained on preprocessed MRI data, with the same image processing steps applied as for X-rays (normalization, resizing, and data augmentation). This YOLO model proved to be successful in detecting KOA-related features in an MRI image and was selected as the final model for KOA prediction in MRI images.

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Model Training Process

The model training process was the same for X-ray and MRI. Datasets were first split into training, validation and testing sets to assess model performance. The models automatically extracted features and classified the pre-processed images. Training involved data augmentation (rotation, flipping) to diversify the dataset and enhance the model's ability to generalize. Learning rate, batch size and training epochs were fine-tuned experimentally for optimal results. Models were trained iteratively with validation on separate datasets to detect overfitting. Models were compared based on performance metrics and reproducibility.

Combination of Predictions

Once the X-ray and MRI models were trained separately, the predictions from the two models were to arrive at the final prediction. Both models provide a classification output (KOA or not) and respective confidence scores. X-ray models offer insights into structural degeneration related to bone, while MRI models focus on soft tissues. The bimodal approach helps improve the prediction accuracy and precision of single-modality approaches.

Explainable AI Integration

To enhance the system's interpretability, explainable AI methods were applied. Explanation techniques in the form of visualizations (heatmaps) were created to show critical regions in the input images used for prediction. This enables clinicians to see which regions of the knee joint were important in the classification, thus improving trust. The inclusion of explainability allows the system to provide not only diagnostic predictions but also enable clinical confirmation.

Output and Interface

The user-friendly interface is designed for users to upload images and obtain predictions. The results include classification (KOA-positive KOA-negative), confidence scores and visualizations. This integration allows the models to be easily integrated within a broader health system or as a standalone interpretable diagnostic tool.

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2.3.2 Testing

Testing is an integral part of ensuring the accuracy, robustness, and performance of the proposed dual-modality KOA prediction system. Testing is essential to make sure that the system behaves as intended in various scenarios and delivers reliable results. Testing was completed using functional, performance and integration testing.

Functional Testing

Functional testing was undertaken to ensure that the various functions of the system are correctly implemented. This involved testing that the system is able to take input images, run predictions and produce the output without any errors.

First, the input image processing was verified by loading X-ray and MRI images in various formats and resolutions. The system correctly verified the data and checked its compatibility with the model. The image preprocessing module was then tested to ensure that noise reduction, normalization, ROI cropping and resizing were performed accurately on each image. The deep learning predictions module was then tested by feeding both X-ray and MRI images to the trained YOLO models. The system accurately produced labels that predicted KOA-positive and KOA-negative classes. Several test cases were tested to check the prediction consistency on multiple images.

Further, output generation features were verified to ensure proper display of the results, such as class labels and associated probabilities. The explainable AI feature was also verified by producing visual explanations, like heatmaps, to ensure that the most relevant parts of the knee joint were highlighted.

Performance Testing

Performance testing was conducted to assess the speed, accuracy and reliability of the models. This involved evaluating the system's performance in a real-world scenario using a test dataset. The efficiency of the models was assessed by standard classification metrics (accuracy, precision, recall, F1-score, and ROC-AUC), giving an overview of the overall performance of the models. Particular care was taken to reduce false negatives, as the consequences of failing to identify a KOA-positive case might be a delay in diagnosis and treatment.

The models based on YOLO, chosen as the final implementation for X-ray and MRI, showed good predictive accuracy and generalization. The YOLO models outperformed earlier CNN models (such as LeNet-5, AlexNet and GoogLeNet) in terms of feature detection and image classification. Besides accuracy, the time taken for inference was also assessed to determine the efficiency of predictions. GPU-acceleration was used to increase the speed of This system, allowing for predictions in

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quasi real-time. This is crucial in clinical settings where timely decisions are needed. Additionally, the models' robustness was assessed by testing them on various sub-datasets to ensure that the system provides consistent predictions under different image conditions and qualities.

Integration Testing

Integration testing was conducted to ensure the various parts of the KOA prediction module integrate properly. The main aim was to test interaction between the preprocessing, model inference, and generation of results. Integration between the image pipeline and YOLO models was carried out to verify the images successfully reached the models after preprocessing without any data issues. The pipeline ensured data integrity was preserved.

We also tested the integration of multimodal predictions to ensure that the X-ray and MRI models produced complementary predictions. The system was shown to be able to process each modality separately and then combine them to generate a consolidated prediction - enhancing confidence. Additionally, the explainable AI component was integrated into the prediction system to provide visual explanations for the system's prediction. This was tested with various inputs to ensure that the explanations were consistent with the predictions made by the model. The system was also integrated with the entire application environment, including the user interface and data storage, to verify compatibility. This verified the system's integration, with all modules working seamlessly.

3 Result & Discussion

3.1 Result

The performance of the proposed dual-modality KOA detection system was evaluated using both X-ray and MRI datasets. The evaluation was conducted using standard classification metrics including accuracy, precision, recall, F1-score, and ROC-AUC. In addition, training behavior, confusion matrices, and explainable AI (XAI) outputs were analyzed to assess model performance and interpretability.

X-ray Model Results

The YOLO-based model developed for X-ray image classification demonstrated strong performance in detecting Knee Osteoarthritis. As shown in the validation accuracy plot, the model achieved a steady improvement during training, increasing from approximately 68% in the initial epochs to over 91% after convergence. The

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validation accuracy stabilized after multiple epochs, indicating that the model successfully learned meaningful features without overfitting.

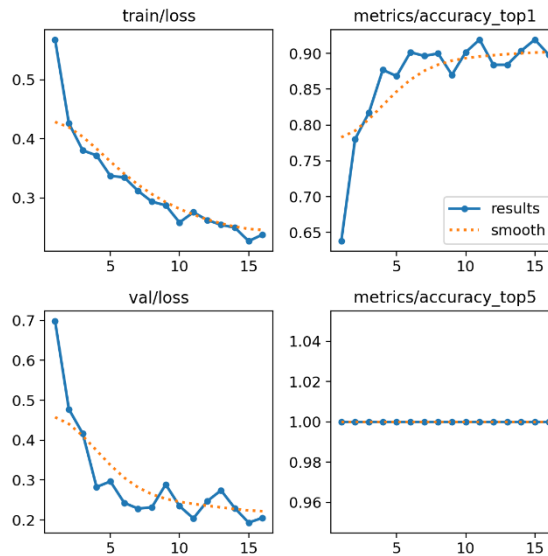


Figure 8 X-ray YOLOv8 Model Plots

The training and validation loss curves show a consistent decrease throughout the training process, with both curves converging smoothly. This indicates effective optimization and stable learning behavior. The absence of large divergence between training and validation loss suggests that the model generalizes well to unseen data.

According to the classification report, the model achieved a validation accuracy of 91.04% and a test accuracy of 90.83%, with both classes (normal and osteoarthritis) demonstrating balanced precision and recall values. The ROC-AUC values of 97.25% for validation and 97.46% for test further confirm the model's strong discriminative capability.

```
=== Validation (YOLOv8, th=0.50) ===
           precision    recall  f1-score   support

   Normal           0.88      0.91      0.89       238
 Osteoarthritis      0.93      0.91      0.92       331

   accuracy                   0.91       569
  macro avg           0.91      0.91      0.91       569
 weighted avg           0.91      0.91      0.91       569

Validation ROC-AUC: 97.25%
Validation Accuracy: 91.04%
```

Figure 9 X-ray Yolo Model Validation Classification Report

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```
=== Test (new stratified) (YOLOv8, th=0.50) ===
              precision    recall  f1-score   support

   Normal           0.87       0.92       0.89         239
 Osteoarthritis       0.94       0.90       0.92         328

 accuracy              0.91         567
 macro avg           0.90       0.91       0.91         567
 weighted avg        0.91       0.91       0.91         567

Test (new stratified) ROC-AUC: 97.46%
Test (new stratified) Accuracy: 90.83%
```

Figure 10 X-ray Yolo Model Test Classification Report

The confusion matrix analysis shows that the model correctly classified a large number of samples in both classes, with relatively low misclassification rates. The number of false negatives is minimal, which is particularly important in medical diagnosis, as it reduces the risk of missing KOA-positive cases.

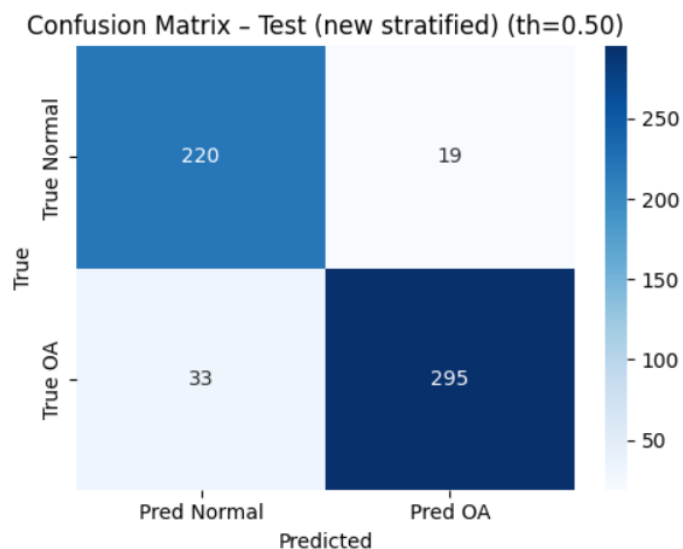


Figure 11 XRay Yolo Model Test Confusion Matrix

MRI Model Results

The YOLO-based model developed for MRI image classification also demonstrated reliable performance, although slightly lower compared to the X-ray model due to the complexity of MRI data.

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The validation accuracy increased progressively from approximately 68% to around 83% across training epochs, indicating stable learning. The training and validation loss curves show a consistent downward trend, demonstrating proper convergence of the model.

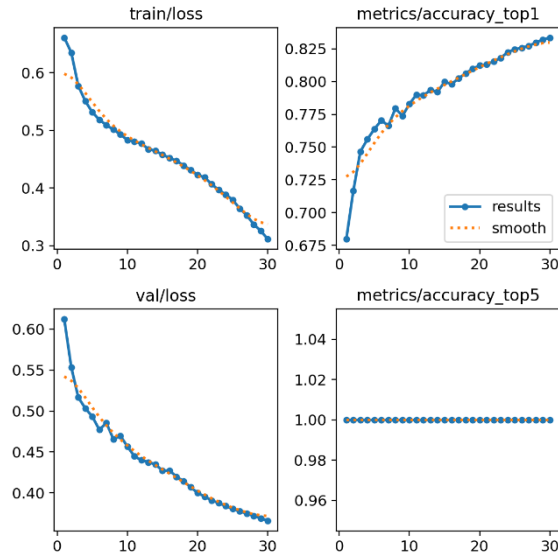


Figure 12 MRI YOLOv8 Model Plots

The classification results indicate a validation accuracy of 83.31% and a test accuracy of 83.98%, with balanced performance across both normal and abnormal classes. The ROC-AUC values of 91.60% (validation) and 92.02% (test) confirm that the model is capable of effectively distinguishing between KOA and non-KOA cases in MRI images.

```
=== Validation (YOLOv8, th=0.40) ===
      precision    recall  f1-score   support

   normal         0.85      0.85      0.85     4977
  abnormal         0.81      0.81      0.81     3951

   accuracy              0.83     8928
  macro avg         0.83      0.83      0.83     8928
 weighted avg         0.83      0.83      0.83     8928

Validation ROC-AUC: 91.60%
Validation Accuracy: 83.31%
```

Figure 13 MRI Yolo Model Validation Classification Report

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```

=== Test (new stratified) (YOLOv8, th=0.40) ===
      precision    recall  f1-score   support

   normal         0.86      0.86      0.86     4977
  abnormal         0.82      0.82      0.82     3951

   accuracy              0.84     8928
  macro avg         0.84      0.84      0.84     8928
 weighted avg         0.84      0.84      0.84     8928

Test (new stratified) ROC-AUC: 92.02%
Test (new stratified) Accuracy: 83.98%

```

Figure 14 MRI Yolo Model Test Classification Report

The confusion matrix shows that the model successfully identifies the majority of cases, although there is a slightly higher misclassification rate compared to the X-ray model. This is expected due to the higher variability and complexity of MRI images, which require more detailed feature extraction.

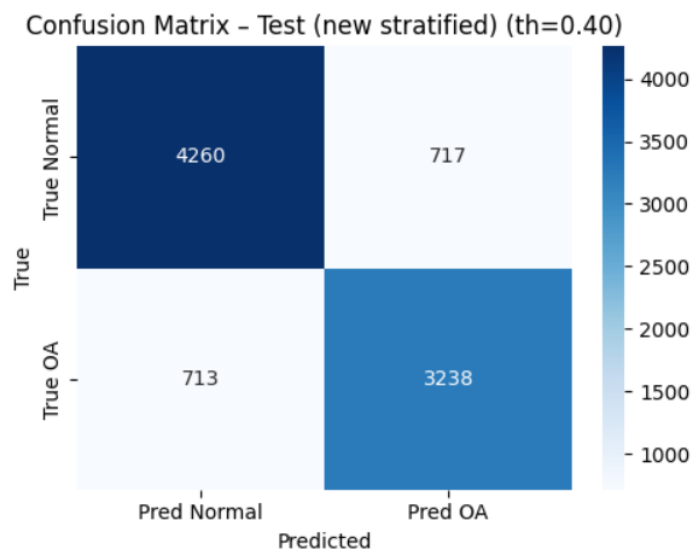


Figure 15 MRI Yolo Model Test Confusion Matrix

Comparative Analysis of X-ray and MRI Models

A comparison of the two models highlights that the X-ray-based model achieved higher overall accuracy and ROC-AUC scores, indicating stronger performance in classification tasks. This can be attributed to the relatively clearer structural patterns present in X-ray images.

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However, the MRI model provides valuable complementary information by capturing soft tissue characteristics that are not visible in X-ray images. While its standalone accuracy is slightly lower, it contributes important diagnostic insights that enhance the overall system capability when combined with X-ray predictions.

This validates the effectiveness of the dual-modality approach, where both models together provide a more comprehensive understanding of KOA compared to using a single modality.

Training Behavior and Model Stability

The training curves for both X-ray and MRI models demonstrate stable learning behavior. In both cases, validation accuracy improves consistently while loss decreases steadily, indicating proper optimization of model parameters. No significant overfitting was observed, as the gap between training and validation performance remained minimal. The models were able to generalize well to unseen test data, as reflected in the close alignment between validation and test accuracies.

Explainable AI (XAI) Results

To enhance interpretability, explainable AI techniques including Grad-CAM, LIME, and SHAP were applied to both X-ray and MRI models. The Grad-CAM visualizations highlight the regions within the knee joint that contributed most to the model's predictions. In X-ray images, the model focuses on areas around the joint space and bone structures, which are clinically relevant indicators of KOA. Similarly, in MRI images, Grad-CAM highlights soft tissue regions, including cartilage and surrounding structures.

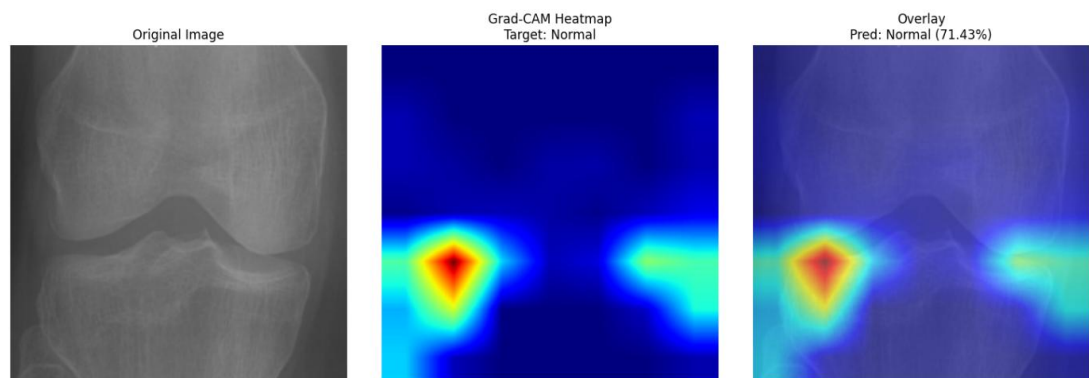


Figure 16 Grad-CAM XAI output for X-ray model

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LIME explanations provide region-based segmentation of the images, identifying specific areas that influence the prediction outcome. These segmented regions correspond to medically relevant features, confirming that the model is learning meaningful patterns rather than irrelevant noise.

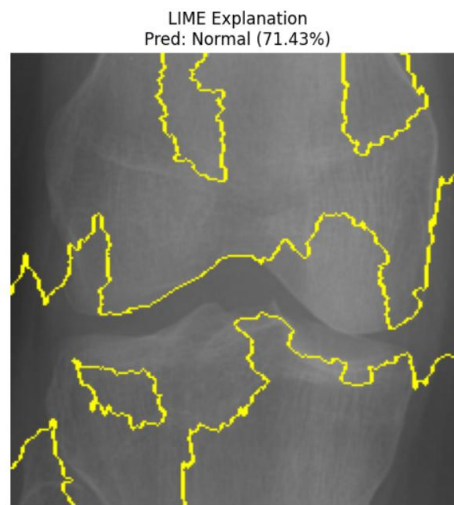


Figure 17 LIME XAI output for X-ray model

SHAP visualizations further quantify the contribution of different regions in the image, showing both positive and negative influences on the prediction. These results demonstrate that the model's decision-making process is interpretable and aligned with clinical understanding.

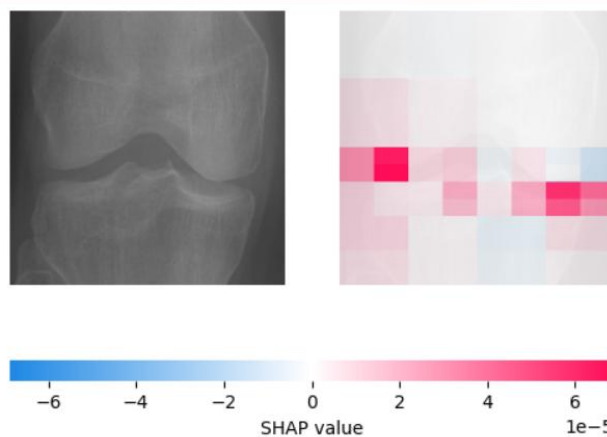


Figure 18 SHAP XAI output for X-ray model

The integration of XAI techniques significantly improves transparency, allowing clinicians to verify and trust the model's predictions. This is a critical requirement for real-world deployment in healthcare environments.

3.2 Research Findings

The experimental study of the proposed dual-modality KOA detection system offers several significant insights that are valuable to both academic studies and clinical applications. First, the research demonstrates that the deep learning-based YOLO models are very efficient for KOA detection using medical images. The X-ray-based model attained a validation accuracy of more than 91% and a ROC-AUC of more than 97%, reflecting its excellent classification ability and discriminative ability. The MRI model also obtained acceptable results with validation accuracy and test accuracy greater than 83% and ROC-AUC greater than 91%, respectively. These findings suggest that the object detection-oriented YOLO can be effectively applied to medical image classification. Its capacity to preserve spatial information and the detection of key regions is beneficial for identifying disease-specific patterns in X-ray and MRI images.

The results demonstrate that the X-ray model achieved higher accuracy and stability of classification results compared to the MRI model. This indicates that X-rays are more likely to capture KOA-related structures, such as joint space narrowing and bone abnormalities, and these features can be readily learned by deep learning models. The strong scores for precision and recall in the X-ray model also suggest that it is capable of accurately identifying KOA. X-rays may therefore serve as a gold standard for KOA diagnosis.

While the MRI-based model was marginally worse at predicting KOA than the X-ray-based model, it gave us important information about soft tissue changes not apparent in X-rays. The MRI-based predictions showed that the model was able to identify cartilage degradation and other early signs of KOA. This underscores the value of MRI as a complementary imaging technique. This research supports that using both MRI and X-rays can enhance diagnostic accuracy, as features of both hard and soft tissues are captured.

The confusion matrix analysis reveals that the models have a low number of false negatives, especially in the X-ray model. This is particularly important because false negatives in medical diagnosis can result in a missed diagnosis and subsequent treatment. The fact that the system accurately classifies most of the KOA-positive cases shows that it can be used as an effective decision support system for early detection.

One of the key contributions of this study is the validation of the two-modal approach. The dual-modality approach allows the system to combine useful information to learn from X-ray and MRI models, overcoming shortcomings of the unimodal models identified in the research gap. This multi-modal approach boosts the stability and accuracy of the system, making it more feasible for practical use in the clinic. This proves that multi-modal learning is a promising avenue for developing medical diagnosis systems.

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Additionally, This research demonstrates the utility of explainable AI for enhancing model interpretability. Grad-CAM, LIME and SHAP techniques provide clear visual and quantitative explanations for model predictions.

These methods showed that the models are attending to the relevant regions of knee joints, such as joint spaces in X-rays and soft tissues in MRI scans. This agreement with clinical expertise assures us that the models are focusing on relevant features, as opposed to spurious patterns. The use of XAI not only promotes model transparency but also boosts clinician trust in the diagnosis of the AI systems, contributing to their adoption in clinical practice.

3.3 Discussion

The findings from the proposed dual-modality approach for KOA detection highlight the potential of deep learning in medical imaging, and also offer valuable insights into the advantages and challenges of various types of models and imaging techniques. A key insight from this research is the better performance of the X-ray-based YOLO model over the MRI model. This is due to the fact that X-rays can detect structural changes in KOA patients, such as joint space narrowing and osteophyte formation. These are more easily detected by deep learning models, leading to more accurate classification and better model stability. This result is in line with the current research, where X-ray-based methods have demonstrated superior performance to detect KOA as a result of their representation of bone-related abnormalities.

On the other hand, the MRI-based approach, while slightly less accurate, is able to delve deeper into soft tissue structures, like cartilage and ligaments. The potential lesser performance of the MRI model is due to greater complexity in MRI images. MRI images have more nuanced features than X-rays, and more sophisticated feature extraction techniques and larger training sets may be needed for effective learning. Nonetheless, the MRI model performed well, suggesting its ability to diagnose early onset KOA that may not have obvious structural changes on X-rays.

This research supports the use of a multimodal approach. Unimodal systems can work with only one type of information, whereas the integration of X-ray and MRI allows the system to incorporate structural and soft tissue information about the disease. This overcomes the key research gap highlighted in previously published works, which report mostly unimodal models that are insufficient to deliver a holistic diagnostic system. The combination of modalities makes the system more robust and effective in clinical practice. Another key contribution of this study is the use of YOLO-based models for medical image classification. YOLO algorithms were originally designed for object detection, and showed good performance in localizing regions and extracting features from X-rays and MRIs. YOLO's ability

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to quickly infer and locate features makes it a good candidate for real-time applications in medicine. This indicates that contemporary object detection models can perform better than conventional CNN-based models in medical imaging tasks.

The use of explainable AI is a crucial step towards solving the black-box problem. The Grad-CAM, LIME and SHAP explanations showed that the models attended to medically significant regions for their predictions. This congruence between model predictions and medical understanding is essential to earn the trust of clinicians. Furthermore, the visual explanations enable clinicians to interpret the predictions and consider the system as a support system rather than a decision maker.

While the proposed system exhibits promising performance, there are some limitations. The accuracy of the MRI model suggests there is room for improvement, by utilizing larger and more diverse datasets, and more sophisticated model architectures optimized for volumetric data. Similarly, while using two modalities enhances performance, the current system processes modalities in isolation, then merges them. This could be improved with more sophisticated fusion of features at earlier layers of the model.

Moreover, the approach needs substantial validation using a variety of patient populations and imaging scenarios for clinical implementation. Imaging equipment, data quality, and patient demographic can play a role in the performance of the system. Hence, further testing is required for widespread use.

4. Summary of Individual Contributions – Fernando W.D.A (IT22223708)

This component focuses on the development of a dual-modality deep learning system for Knee Osteoarthritis (KOA) detection using X-ray and MRI images. The work includes data preprocessing, model experimentation, and implementation of multiple deep learning architectures such as LeNet-5, AlexNet, GoogLeNet, and YOLO-based models. Following comparative analysis, YOLO-based models were selected as the final approach for both X-ray and MRI image classification due to their superior performance and efficiency. A complete image preprocessing pipeline, including normalization, enhancement, and region of interest extraction, was implemented to improve model accuracy.

In addition, explainable AI techniques such as Grad-CAM, LIME, and SHAP were integrated to enhance model interpretability and provide visual explanations for predictions. The component also includes model evaluation using standard performance metrics and contributes to the overall system by providing accurate and reliable KOA detection results.

5. Conclusion

This research presented a comprehensive multi-modal approach for the detection of Knee Osteoarthritis (KOA) by integrating deep learning techniques with medical imaging data. The primary objective of this study was to develop an accurate and reliable system capable of identifying KOA using both X-ray and MRI images, thereby addressing the limitations of traditional unimodal diagnostic methods. The implementation of YOLO-based deep learning models for both X-ray and MRI image analysis proved to be highly effective. The X-ray model achieved strong performance with high accuracy and excellent ROC-AUC values, demonstrating its capability in detecting structural abnormalities associated with KOA. The MRI model, although comparatively lower in accuracy, successfully captured soft tissue-related features, contributing valuable insights for early-stage disease detection. These findings validate the importance of leveraging multiple imaging modalities to achieve a more comprehensive understanding of KOA.

A key contribution of this research is the development of a dual-modality framework, which combines the strengths of both X-ray and MRI imaging. This approach addresses the research gap identified in existing studies, where most systems rely on a single modality and fail to capture the full complexity of the disease. By integrating predictions from both models, the system enhances diagnostic accuracy and reliability, making it more suitable for real-world clinical applications. In addition, the incorporation of Explainable AI (XAI) techniques, including Grad-CAM, LIME, and SHAP, significantly improves the interpretability of the system. These techniques provide visual and quantitative explanations for model predictions, allowing clinicians to understand the reasoning behind automated decisions. This transparency is critical for building trust and supporting the adoption of AI-based systems in healthcare environments.

The results of this study demonstrate that the proposed system is capable of achieving high performance, stable learning behavior, and effective generalization across datasets. Furthermore, the system is computationally efficient and scalable, making it suitable for integration into clinical workflows and healthcare systems. Despite its strengths, this research also identifies areas for future improvement. Enhancing the performance of the MRI model through larger and more diverse datasets, as well as exploring advanced architectures and deeper multi-modal fusion techniques, could further improve system accuracy. Additionally, real-world clinical validation and deployment remain essential steps for ensuring the practical applicability of the system.

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7. GLOSSARY

Knee Osteoarthritis (KOA)

A chronic degenerative joint disease characterized by the breakdown of cartilage in the knee, leading to pain, stiffness, and reduced mobility.

Cartilage

A smooth, rubber-like tissue that covers the ends of bones in joints, allowing them to move easily without friction.

X-ray Imaging

A medical imaging technique used to visualize bone structures, commonly used to detect joint space narrowing and bone abnormalities in KOA.

Magnetic Resonance Imaging (MRI)

An advanced imaging technique that provides detailed images of both bone and soft tissues, including cartilage, ligaments, and muscles.

Kellgren–Lawrence (KL) Grading System

A widely used radiographic classification system that categorizes the severity of osteoarthritis from Grade 0 (normal) to Grade 4 (severe).

Deep Learning

A subset of machine learning that uses neural networks with multiple layers to automatically learn and extract features from data.

Convolutional Neural Network (CNN)

A type of deep learning model specifically designed for image analysis, capable of detecting patterns and features within images.

YOLO (You Only Look Once)

A real-time object detection algorithm that can identify and classify objects within an image efficiently. In this research, it is adapted for KOA detection.

Multi-Modal Learning

An approach that combines multiple types of data (e.g., X-ray and MRI images) to improve model performance and accuracy.

Explainable Artificial Intelligence (XAI)

Techniques that provide transparency in AI models by explaining how predictions are made, improving trust and interpretability.

Grad-CAM (Gradient-weighted Class Activation Mapping)

A visualization technique that highlights important regions in an image that influence the model's prediction.

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LIME (Local Interpretable Model-agnostic Explanations)

A method that explains individual predictions by approximating the model locally and identifying important input regions.

SHAP (SHapley Additive exPlanations)

A method that explains model predictions by assigning contribution values to each feature based on game theory.

Feature Extraction

The process of automatically identifying important patterns or characteristics from data, especially images.

Classification

A machine learning task that involves categorizing input data into predefined classes, such as KOA-positive or KOA-negative.

ROC-AUC (Receiver Operating Characteristic – Area Under Curve)

A performance metric used to evaluate classification models, indicating the model's ability to distinguish between classes.

Precision

The proportion of correctly predicted positive observations out of all predicted positives.

Recall (Sensitivity)

The proportion of correctly identified positive cases out of all actual positive cases.

F1-Score

A performance metric that combines precision and recall into a single value to evaluate model accuracy.

Overfitting

A condition where a model performs well on training data but poorly on unseen data due to excessive learning of noise.

Data Augmentation

Techniques used to artificially increase dataset size by applying transformations such as rotation, flipping, and scaling.